

Supplementary Document S1 — Amendment 01

Type: Post-hoc methodological amendment to the pre-registered analysis plan

(S1) **Frozen timestamp:** 2026-05-21 21:49 KST (Bisque substitution decision

recorded *immediately* upon documented CIBERSORTx server unavailability)

Parent document: manuscript/S1_Pre-registered_Analysis_Plan.md (frozen

2026-05-21) **Amendment author:** Bang Changseok **Status:** Recorded prior to

Bisque execution. The pre-registered Spearman $r \geq 0.7$ threshold (S1 §4-4) is

preserved without modification.

1. Summary

Section §4-3 of the parent S1 document specified **CIBERSORTx (Stanford web server, S-**

mode batch correction) as the cross-validation method for the MuSiC-derived cell

fractions. This amendment substitutes **Bisque (R package BisqueRNA, Jew et al. 2020**

Nature Communications) as a functionally equivalent reference-based deconvolution

method, with all other aspects of the analytical commitment preserved.

The pre-registered decision rule — that the MuSiC cell fractions must show Spearman $r \geq$

0.7 agreement with the cross-validation method's estimates for partial-correlation results

to be reported as the main finding (S1 §4-4) — is **carried over verbatim** to the Bisque

comparison.

2. Reason for amendment

On 2026-05-21 at 21:49 KST, the CIBERSORTx server at <https://cibersortx.stanford.edu> was confirmed unavailable. Server status was independently verified at <https://downforeveryoneorjustme.com/>. The status was not transient — repeated checks over the preceding hour returned the same failure. Continued waiting for server restoration would introduce an indefinite analytical delay incompatible with the project timeline.

Bisque (Jew *et al.* 2020 *Nature Communications* 11:1971) was selected because:

1. **Same conceptual approach:** reference-based deconvolution of bulk RNA-Seq using single-cell reference data with marker-gene selection and proportion estimation.
2. **Functionally equivalent inputs/outputs:** takes a single-cell reference (cell type labels + subject identifiers) and a bulk mixture, returns per-sample cell-type proportions.
3. **Independent implementation and statistical model:** Bisque uses a marker-gene-based decomposition with cell-type-specific reference profiles weighted by within-subject variance, which is methodologically distinct from MuSiC's hierarchical weighted non-negative least squares. Both methods are designed to address the same problem (cell composition inference) using different statistical formulations, making them appropriate independent cross-validation partners.

- 40 4. **Local execution:** as an R package, Bisque eliminates dependence on an external
41 web server, increasing reproducibility.
- 42 5. **Published cross-comparisons:** Bisque has been benchmarked against MuSiC,
43 CIBERSORTx, and other deconvolution methods in the original publication (Jew *et al.*
44 2020) and shows comparable accuracy on reference cohorts.

45 3. Specific changes

46 Parent document S1, §4-3 — Pre-specified data sources for Stage 2

47 **Original text** (frozen 2026-05-21):

48 “**Deconvolution method:** MuSiC v1+ (Wang *et al.* 2019 *Nat Commun*) with
49 Smillie scRNA as reference (broad cell types as `clusters`, donor × condition as
50 `subjectID`). **CIBERSORTx (Stanford web server, S-mode batch correction) will**
51 **be used as a cross-validation method for the cell fractions if MuSiC results**
52 **meet criterion in §4-4.”**

53 **Amended text** (this amendment, 2026-05-21 21:49 KST):

54 “**Deconvolution method:** MuSiC v1+ (Wang *et al.* 2019 *Nat Commun*) with
55 Smillie scRNA as reference (broad cell types as `clusters`, donor × condition as
56 `subjectID`). **Bisque (R package BisqueRNA; Jew *et al.* 2020 *Nat Commun***
57 **11:1971) using `ReferenceBasedDecomposition()` with `use.overlap = FALSE`**

will be used as a cross-validation method for the cell fractions if MuSiC results meet criterion in §4-4.”

Parent document S1, §4-4 — Pre-specified Stage 2 decision rules

No change. The decision rule “MuSiC cell fractions will be cross-validated against [cross-validation method] estimates. Reasonable agreement (Spearman $r \geq 0.7$ across samples for the dominant cell type) is required for partial-correlation results to be reported as the main finding” is preserved exactly as originally specified. Only the identity of the cross-validation method changes.

4. What this amendment does NOT change

- Pre-registered Spearman $r \geq 0.7$ threshold (§4-4) — preserved
- Single-cell reference data (Smillie GSE116222 broad cell types) — same
- Bulk mixture (GSE193677, $n=2,483$ QC-passed samples) — same
- Broad cell-type categories (Epithelial / Immune / Stromal / Other) — same
- Subject identifier scheme (donor $\times$ condition) — same
- Reporting standards for downstream partial-correlation analyses — same
- All Stage 1 results (S1 §2-3) — already executed, not re-touched
- Stage 2 single-cell analysis (Smillie scRNA, within-cell-type correlations) — not affected
- External validation cohort (GSE235236) — not affected

- Pseudobulk-per-donor robustness check — not affected
- Bootstrap 95% CI computation for partial Spearman — not affected
- Random-effects meta-analysis (metafor REML) — not affected

5. Pre-commitment to reporting

Regardless of the outcome of the Bisque cross-validation, this amendment commits to transparent reporting of:

1. **Both the Spearman and Pearson correlation coefficients** of MuSiC vs. Bisque Epithelial and Immune fraction agreement.
2. **The decision outcome relative to the pre-registered Spearman $r \geq 0.7$ threshold** (PASS / FAIL).
3. **A Bland-Altman style plot** for the Epithelial fraction comparison (mean vs. difference) to assess systematic bias.
4. **The full Bisque cell fraction table** as a Supplementary Data file.
5. **The amendment itself** (this document) as a Supplementary Document accompanying the parent S1.
6. **If the pre-registered threshold is not met**, the partial-correlation analyses will be reported as exploratory rather than as the main finding, per the original §4-4 decision rule.

The amendment is therefore a methodological substitution made under documented external constraint, not a post-hoc relaxation of the pre-registered evidence standard.

6. Citation

When citing the deconvolution methodology in the main manuscript, the following references will be used:

- Wang X, Park J, Susztak K, Zhang NR, Li M. **Bulk tissue cell type deconvolution with multi-subject single-cell expression reference.** *Nat Commun* 2019;10(1):380. (MuSiC, primary method)
- Jew B, Alvarez M, Rahmani E, Miao Z, Ko A, Garske KM, Sul JH, Pietiläinen KH, Pajukanta P, Halperin E. **Accurate estimation of cell composition in bulk expression through robust integration of single-cell information.** *Nat Commun* 2020;11(1):1971. (Bisque, cross-validation)

7. Archive details

- Amendment ID:** S1-A01
- Parent document:** S1_Pre-registered_Analysis_Plan.md (DOI: 10.17605/OSF.IO/AB5MY pending OSF/Zenodo deposition)
- Freeze timestamp:** 2026-05-21 21:49 KST
- Intended deposition:** same OSF/Zenodo entry as parent document, as a versioned amendment file
- Citation in manuscript:** “Following the documented unavailability of the CIBERSORTx Stanford server on 2026-05-21, the cross-validation method was

116 substituted with Bisque (Jew *et al.* 2020) per pre-registered Amendment 01
117 (Supplementary Document S1-A01). The Spearman $r \geq 0.7$ threshold and all other
118 pre-registered commitments were preserved.”

119

120 *End of S1 Amendment 01. This file is the immutable historical record of the methodological*
121 *substitution from CIBERSORTx to Bisque, executed under documented external server*
122 *unavailability with full preservation of the pre-registered evidence standard.*